

Experiment 5

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Title:

Develop a web application to fetch real-time weather data for a city. Use AJAX & a public weather API to retrieve information. Display temperature, humidity, and weather conditions dynamically.

Theory:

A weather web application provides real-time climate information by communicating with external servers using web technologies. In this experiment, AJAX (Asynchronous JavaScript and XML) is used to fetch live weather data from a public Weather API without refreshing the webpage.

The Weather API returns data in JSON format, which contains various weather parameters such as temperature, humidity, and weather description. JavaScript processes this data and updates the webpage dynamically using DOM manipulation.

AJAX improves user experience by allowing asynchronous data transfer between client and server. The user enters a city name, and the application retrieves and displays updated weather information instantly.

Algorithm:

1. Start the experiment.
2. Create an HTML file for user interface.
3. Add an input field to enter the city name.

4. Add a button to fetch weather details.
5. Use JavaScript to capture the city name entered by the user.
6. Send an AJAX request to the Weather API using fetch().
7. Receive weather data in JSON format.
8. Extract temperature, humidity, and weather condition from the response.
9. Display the fetched data dynamically on the webpage.
10. Stop the experiment.

Program :

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
<meta charset="UTF-8">
```

```
<title>Colorful Weather App</title>
```

```
<style>
```

```
* {
```

```
margin: 0;
```

```
padding: 0;
```

```
box-sizing: border-box;
```

```
font-family: 'Segoe UI', sans-serif;
```

```
}
```

```
body {  
    height: 100vh;  
    display: flex;  
    justify-content: center;  
    align-items: center;  
    background: linear-gradient(-45deg, #4facfe, #00f2fe, #43e97b, #38f9d7);  
    background-size: 400% 400%;  
    animation: gradientBG 10s ease infinite;  
}
```

```
@keyframes gradientBG {  
    0% {background-position: 0% 50%;}  
    50% {background-position: 100% 50%;}  
    100% {background-position: 0% 50%;}  
}
```

```
.container {  
    background: rgba(255, 255, 255, 0.2);  
    backdrop-filter: blur(15px);  
    padding: 40px;  
    border-radius: 20px;  
    width: 350px;  
    text-align: center;  
    color: white;
```

```
    box-shadow: 0 10px 25px rgba(0,0,0,0.3);  
}
```

```
h2 {  
    margin-bottom: 20px;  
    font-size: 28px;  
    letter-spacing: 1px;  
}
```

```
input {  
    width: 100%;  
    padding: 12px;  
    border: none;  
    border-radius: 25px;  
    outline: none;  
    margin-bottom: 15px;  
    font-size: 16px;  
    text-align: center;  
}
```

```
button {  
    width: 100%;  
    padding: 12px;  
    border: none;  
    border-radius: 25px;
```

```
background: #ff7eb3;
color: white;
font-size: 16px;
cursor: pointer;
transition: 0.3s;
font-weight: bold;
}
```

```
button:hover {
  background: #ff4f81;
  transform: scale(1.05);
}
```

```
.weather-box {
  margin-top: 20px;
  padding: 15px;
  border-radius: 15px;
  background: rgba(255, 255, 255, 0.25);
  font-size: 16px;
  line-height: 1.6;
}
```

```
.temp {
  font-size: 30px;
  font-weight: bold;
```

```

        margin-bottom: 5px;
    }
</style>
</head>

<body>

<div class="container">

    <h2>☁ Weather App</h2>

    <input type="text" id="city" placeholder="Enter city name">

    <button onclick="getWeather()">Get Weather</button>

    <div id="result"></div>

</div>

<script>

function getWeather() {

    var city = document.getElementById("city").value;

    var apiKey = "YOUR_API_KEY";

    var url = "https://api.openweathermap.org/data/2.5/weather?q="

        + city + "&units=metric&appid=" + apiKey;

    fetch(url)

    .then(response => response.json())

    .then(data => {

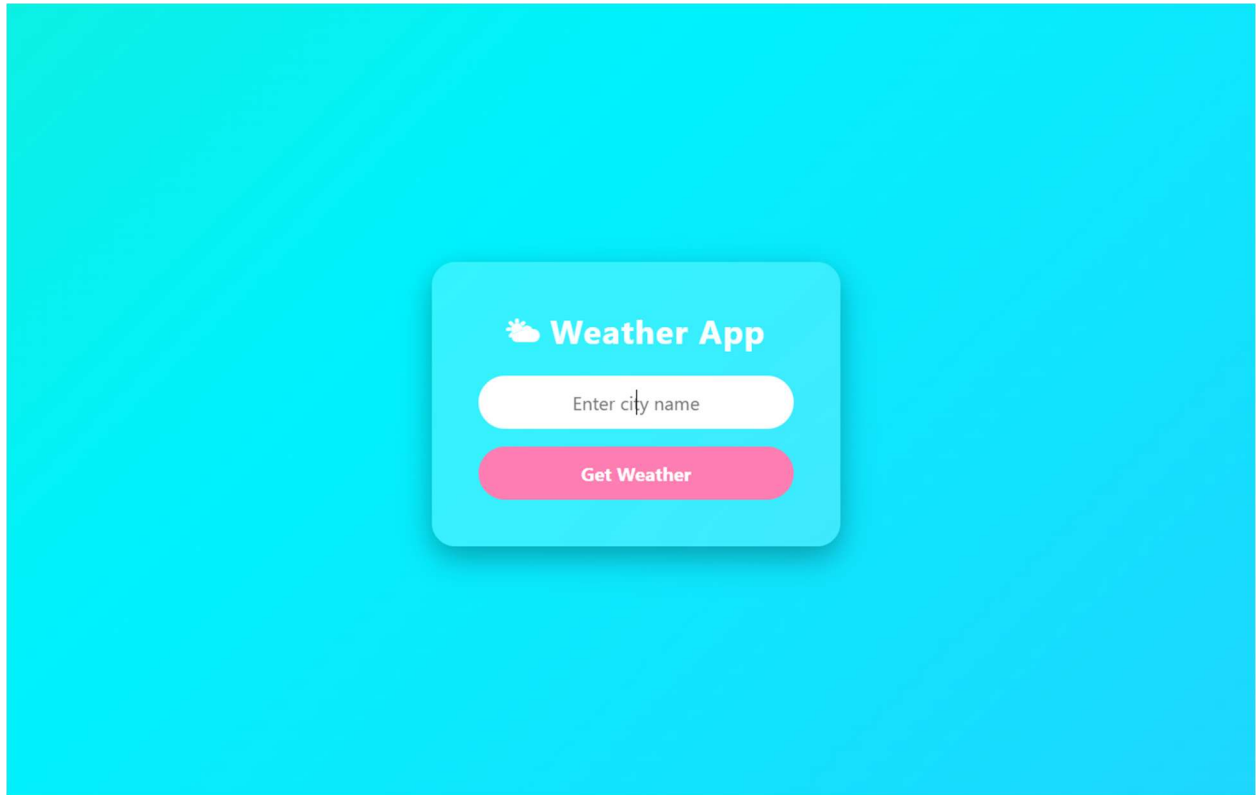
        if (data.cod !== 200) {

```

```
document.getElementById("result").innerHTML =  
    "<div class='weather-box'>City not found!</div>";  
return;  
}
```

```
document.getElementById("result").innerHTML =  
    "<div class='weather-box'>" +  
    "<div class='temp'>" + data.main.temp + "°C</div>" +  
    "Humidity: " + data.main.humidity + "%<br>" +  
    "Condition: " + data.weather[0].description +  
    "</div>";  
})  
.catch(() => {  
    document.getElementById("result").innerHTML =  
        "<div class='weather-box'>Error fetching data!</div>";  
    });  
}  
</script>  
  
</body>  
</html>
```

Output:



Conclusion

Thus, a dynamic weather web application was successfully developed using AJAX and a public Weather API. This experiment demonstrates real-time data fetching, asynchronous communication, and dynamic content updating using JavaScript and DOM manipulation.